

PORTFOLIO BUILDING B25CS0311

ACTIVITY 14 — HACKATHON PREPARATION

TEAM PROJECT: TEAMMATCH

Student Team Formation Platform

1. TEAM INFORMATION

Name	Register Number	Role
T S HARSHAVARDHAN NAYAKA	R25EJ159	Frontend
VIGHNESH P KANHIRAKANDI	R25EJ174	Backend
TERRANCE PAUL S	R25EJ162	All-rounder

Team Objective

The team was formed with complementary responsibilities covering frontend development, backend development, database handling, integration, testing, documentation, and presentation.

2. SELECTED HACKATHON

Innohacks 4.0

Hackathon: Innohacks 4.0

Platform: Devfolio

Location: Ghaziabad, India

Team Size: 2–4 members

Registration Cost: Free

Official Hackathon Link:

<https://innohacks-4.devfolio.co/>

Hackathon Schedule

- **Idea PPT Submission Round:** 10 September 2026 – 26 September 2026
- **Offline Hackathon:** 3–4 October 2026
- **Shortlisting:** Top 40 teams proceed to the offline hackathon.

The hackathon provides multiple technology directions, including AI, blockchain, fintech, AR/VR and AIoT. The team-size requirement of 2–4 members matches our three-member team.

3. BRAINSTORMING — THREE PROBLEM STATEMENTS

IDEA 1 — CampusFix

Problem

Students frequently encounter campus problems such as damaged classroom equipment, cleanliness issues, water problems, Wi-Fi issues, and other maintenance concerns. These problems may be reported informally, making it difficult for students to track whether their complaint has been received, assigned, or resolved.

Proposed Solution

CampusFix would be a web-based campus issue reporting and tracking platform. Students could submit an issue with its category, location, description, and optional image. The concerned department could view the issue and update its status through stages such as:

Reported → Assigned → In Progress → Resolved

Feasibility

High

Potential Impact

High

IDEA 2 — TeamMatch

Problem

Students participating in hackathons, technical competitions, and college projects often find it difficult to identify suitable teammates with complementary skills. A student may need a frontend developer, backend developer, designer, AI/ML member, or presentation member but may not know whom to approach.

Proposed Solution

TeamMatch would be a web-based platform where students create profiles containing their skills, interests, preferred roles, and availability. Students could create project or hackathon requirements and discover potential teammates based on required skills and roles.

Feasibility

High

Potential Impact

High

IDEA 3 — QueueLess Campus

Problem

Students often have to wait physically in queues for administrative services, certificates, appointments, faculty interactions, and other college services. They may not know their expected waiting time or current position in the queue.

Proposed Solution

QueueLess Campus would provide a digital appointment and queue management system. Students could select a service, book a slot or token, and track their queue position. Staff could manage appointments and update the status of requests through a dashboard.

Feasibility

High

Potential Impact

Medium–High

4. PROBLEM IDEA EVALUATION

Criteria	CampusFix	TeamMatch	QueueLess Campus
Feasibility	High	High	High
Student Relevance	High	High	Medium–High
Prototype Clarity	High	High	High
Scope for Future Development	High	High	Medium–High

5. SELECTED PROBLEM

TEAMMATCH — STUDENT TEAM FORMATION PLATFORM

Why this problem was selected

TeamMatch addresses a common challenge faced by students who participate in hackathons, technical competitions, academic projects, and collaborative activities.

The idea has a clear and manageable workflow:

Student Registration → Profile Creation → Project/Hackathon Requirement → Skill & Role Search → Matching → Team Request → Team Formation

The solution can be developed as a functional web prototype without requiring complex machine-learning infrastructure. A simple rule-based matching mechanism can initially compare required skills and roles with student profiles.

The concept also provides a clear division of work among the three team members and offers scope for future improvements such as recommendation systems, notifications, advanced matching, and collaboration features.

6. FINAL PROBLEM STATEMENT

Students often struggle to find suitable teammates for hackathons, college projects, and technical competitions, especially when they need people with specific skills or roles such as

frontend, backend, AI/ML, design, or presentation. TeamMatch is a web-based platform that helps students create profiles containing their skills, interests, preferred roles, and availability, while allowing them to create or search for project and hackathon requirements. The platform matches students with suitable teammates based on required skills and roles, making team formation easier and more organized.

7. PROPOSED SOLUTION

TeamMatch will provide a centralized platform for students to discover and form suitable teams.

Main Features

1. Student Registration

Students can create an account and provide basic information required for team formation.

2. Student Profile

Each student can maintain a profile containing:

- Skills
- Technical interests
- Preferred role
- Areas of interest
- Availability
- Previous project/hackathon interests

3. Project / Hackathon Creation

Students can create a project or hackathon requirement and specify:

- Project title
- Description
- Required skills
- Required roles
- Number of members needed

4. Skill-Based Discovery

Students can search for potential teammates using skills, roles, and interests.

5. Basic Matching

The system compares project requirements with student profile information and displays suitable potential teammates.

6. Team Requests

Students can send team requests and accept or reject incoming requests.

7. Team Dashboard

Once a team is formed, members can view:

- Team members
 - Project information
 - Roles
 - Pending requests
 - Team status
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8. TECHNOLOGY STACK

Component	Technology	Purpose
Frontend	HTML, CSS, JavaScript / React	User interface, forms, pages and dashboards
Backend	Python Flask	Application logic, APIs, authentication and request handling
Database	MySQL	Store users, skills, projects, teams and requests
Matching	Rule-based matching	Compare required skills/roles with student profiles
Version Control	Git & GitHub	Collaborative development and version management

9. IMPLEMENTATION APPROACH

The proposed implementation will follow these stages:

Phase 1 — Planning

Define the user flow, features, roles, and database structure.

Phase 2 — User Management

Implement student registration, login, and profile creation.

Phase 3 — Project Management

Allow students to create and manage hackathon/project requirements.

Phase 4 — Discovery

Implement skill-based search and filtering for potential teammates.

Phase 5 — Matching

Implement a simple rules-based matching mechanism using skills and preferred roles.

Phase 6 — Team Formation

Implement team requests, acceptance/rejection, and team-member management.

Phase 7 — Dashboard

Create a basic dashboard showing project and team information.

Phase 8 — Integration and Testing

Connect the frontend, backend, and database and test the main user flows.

10. ROLE DISTRIBUTION

T S HARSHAVARDHAN NAYAKA — FRONTEND

Responsibilities:

- User interface design
- Registration and profile pages
- Project creation interface
- Search and filtering interface
- Matching results display
- Team dashboard
- Frontend-backend integration support

VIGHNESH P KANHIRAKANDI — BACKEND

Responsibilities:

- Flask backend
- Database structure
- API development
- Authentication
- User and project data handling
- Team request logic
- Backend validation

TERRANCE PAUL S — ALL-ROUNDER

Responsibilities:

- Support frontend and backend integration
 - Matching logic
 - Testing
 - Documentation
 - Debugging support
 - Presentation and demonstration preparation
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11. EXPECTED USER FLOW

REGISTER



CREATE PROFILE



ADD SKILLS & PREFERRED ROLE



CREATE / SEARCH PROJECT



VIEW SUITABLE TEAMMATES



SEND TEAM REQUEST



ACCEPT / REJECT REQUEST



FORM TEAM



MANAGE TEAM THROUGH DASHBOARD

12. EXPECTED OUTCOME

The expected outcome is a functional web-based prototype that simplifies student team formation for hackathons, technical competitions, and academic projects.

The system aims to reduce the difficulty of finding suitable teammates by providing a structured platform where students can clearly communicate their skills, roles, interests, and project requirements.

13. FUTURE SCOPE

The initial prototype can later be extended with:

- Advanced recommendation algorithms
 - AI-assisted teammate recommendations
 - Email or notification alerts
 - Skill verification
 - Student portfolio integration
 - Hackathon discovery
 - Team performance tracking
 - Project collaboration tools
 - Ratings or feedback after project completion
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14. ACTIVITY DELIVERABLE CHECKLIST

Activity Requirement	Status
Team of 2–4 members	✓ Completed

Team roster	✓ Completed
Upcoming hackathon selected	✓ Completed
Hackathon link provided	✓ Completed
Three distinct problem statements	✓ Completed
Feasibility and impact considered	✓ Completed
One problem statement selected	✓ Completed
Final problem statement written	✓ Completed
Technology stack identified	✓ Completed
Implementation approach prepared	✓ Completed
Shared document / GitHub Issue	To be shared with all team members

CONCLUSION

Through this activity, the team completed the initial preparation required before entering the build phase of a hackathon. The team identified an upcoming hackathon, brainstormed multiple real-world problems, evaluated their feasibility and potential impact, and selected TeamMatch as the proposed solution.

The concept provides a practical and scalable approach to helping students find suitable teammates based on skills and roles while giving the team a clear technical direction for future development.